
Style Gravity, in Text: How Long a Human-Authored Poem Opening Survives Inside a Model’s Continuation

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Abstract

Anthropic’s response-prefill affordance let an author impose the opening of a poem as the model’s own in-progress output and have the model simply keep writing. That affordance has since been withdrawn from the entire current Claude model line. We measure what it was worth. We seed poems with deliberately off-distribution openings (skaldic kennings, monosyllabic constraint, lipograms, broken typography) and score each continuation line between two poles estimated inside the same run: the seed’s own style (-1) and that model’s unprompted house style ($+1$), using 24 surface-stylometric features and no embeddings. Across 1,596 generations, five models, four conditions, and fourteen seeds, the pull toward the house style is real and line-by-line measurable, but usually incomplete: 30 of 64 non-control curves never revert within 20 lines. The dominant factor is not the structural channel but a single verbal instruction to sustain the opening’s style: every instructed+sustain curve holds for the full poem, no plain prefill curve does, and plain prefill is the only condition whose mean asymptote lands on the model’s side of zero. On a hard formal constraint (monosyllables) the sustain effect is negative in all eight model $\times$ channel cells with all bootstrap intervals excluding zero; on a register seed (skaldic) it is uneven. Against the visual-domain precedent of autonomous generation loops collapsing onto generic motifs, the within-continuation pull is weaker and interruptible: asking, in words, is what holds the poem where the author put it.

1 Introduction

Anthropic’s API long allowed an author to supply the opening of a response as a trailing assistant turn. The model experienced that text as its own in-progress output and continued it: no instruction, no framing, the author’s lines structurally installed as the model’s voice. This *prefill* affordance has been removed from the current Claude model line (it remains available on the 4.5-generation models used here, but not on Sonnet 5 or Opus 5), a change that occurred between the design of this experiment and its running. What was withdrawn was not a convenience but a specific form of creative agency: the ability to impose one’s own text as the model’s output rather than ask for a continuation of someone else’s. This paper measures what that affordance conferred, what its verbal substitute recovers, and, in the result we did not expect, which of the two is actually the stronger lever.

The instrument is a decay curve. We seed a poem with an opening written deliberately outside anything the model would produce unprompted, let the model continue, and score every continuation line between two poles: the seed’s style and the model’s own. The visual-domain precedent for this kind of measurement is Hintze et al. [1], who drive an autonomous language–image loop for 100 iterations across 700 trajectories and find every run collapsing onto roughly a dozen generic, commercially safe motifs (lighthouses, cathedrals, palatial interiors), robust to model swap and raised

temperature. Their result establishes that a model’s pull toward its own aesthetic centre is measurable and that the destination is a small attractor set. We ask the within-continuation version of the same question, in text: is the pull visible line by line inside a *single* continuation, where no re-encoding step is available to explain the collapse, and how far does it get before the poem ends?

The project was designed around a number: the line at which the model’s house style reclaims the poem. For half of the run, that number does not exist. What the data supports instead is sharper and more useful: seeded style largely persists across 20 lines; the pull toward the model’s register is real in nearly every curve but usually too weak to complete within a poem’s length; and whether the poem holds is decided almost entirely by whether the author *asked*, in words, for the style to be sustained. The verbal request is stronger than the structural affordance it replaces.

2 Related work

Three neighbouring literatures establish that generative models exert a measurable pull toward their own centre, at three different timescales. At the timescale of *training*, recursive training on model-generated data collapses the tails of the content distribution [2]. At the timescale of *use*, co-writing with an instruction-tuned model measurably reduces the diversity of what different authors produce [3], LLM-assisted ideation homogenises ideas across users even as it helps individuals [4], and generative AI raises individual creativity while reducing the collective diversity of novel content [5]. At the timescale of *autonomous iteration*, language–image loops converge onto a small set of generic visual motifs [1]. This paper measures the pull at the shortest timescale of all, line by line within a single continuation, where none of the mechanisms those literatures invoke (retraining, human selection, re-encoding) is available.

On the instrumentation side, our metric descends from computational stylometry: surface features of the kind long used for authorship attribution [6] rather than embeddings, precisely because attribution-style features track *how* a line is written rather than what it is about. The conditions descend from controllable generation: steering by conditioning code or instruction [7], by inference-time attribute control [8], and the decoding-side observation that models have a characteristic register they revert to [9]. Human–AI co-writing systems such as CoAuthor [10] study who contributes what in a shared draft; we study the narrower, mechanistic question of how long the human’s contribution keeps setting the style after the hand-off.

3 Method

Two poles, one metric. “How far has line i drifted from the seed?” has no answer in absolute units, because every continuation moves. The answerable question is which pole line i is closer to. Each line is scored against a centroid of the seed’s lines and a centroid of that model’s unprompted lines:

$$g(i) = \frac{d_{\text{seed}}(i) - d_{\text{baseline}}(i)}{d_{\text{seed}}(i) + d_{\text{baseline}}(i)} \in [-1, +1], \quad (1)$$

where -1 sits exactly on the seeded style, $+1$ on the model’s house style, and 0 is the tipping point. Both poles are estimated inside the same run, in the same feature space, so a model with an idiosyncratic baseline is measured against *its own* baseline: the curve is a per-model steering measurement. The house-style pole comes from the identical poem request with no seed at all (“Write a poem, at least N lines long.”), holding prompt wording constant between seeded and unseeded conditions.

Features. Each line is a vector of 24 surface-stylometric features: syllable and word length, mono-syllable rate, type–token ratio, function-word rate, archaism rate, Latinate-suffix rate, hyphenated compounds, punctuation and dash density, terminal stop versus enjambment, capitalisation, first-person rate, *And*-initial rate, bracket and digit rate, internal space runs, letter-*e* rate, and in-line repetition. Embeddings were rejected on purpose: an embedding model’s own priors would sit between the measurement and the thing measured, and embedding distance is dominated by topic rather than style.

Conditions. Four conditions cross two channels with one instruction. `prefill` supplies the seed as a trailing assistant turn, so the model experiences it as its own output. `instructed` places the

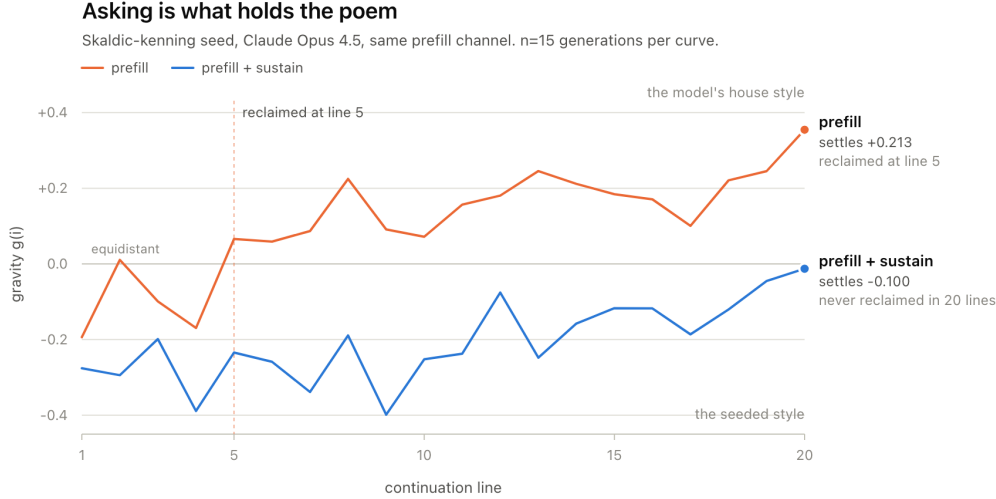


Figure 1: **Asking is what holds the poem.** Line-by-line gravity for a skaldic-kenning seed continued by Claude Opus 4.5 over the same prefill channel, with and without a system-prompt instruction to sustain the opening’s style. -1 sits on the seeded style, $+1$ on the model’s own unprompted style, each pole estimated inside the same run. Without the instruction the poem crosses to the model’s side at line 5 and settles at $+0.213$; with it, the poem never crosses in 20 lines and settles at -0.100 . $n=15$ generations per curve.

seed in the user turn with a bare “continue this” request, a different intervention that is never pooled with prefill. Each has a `+sustain` twin adding a system-prompt instruction to sustain the opening’s diction, syntax, form, and register, deliberately without naming the style (naming it would leak the answer). The contrast between a cell and its `+sustain` twin is the experiment’s central measurement: what a verbal request recovers of what the structural affordance provided.

Design and seeds. 1,596 generations across five models (Claude Haiku 4.5, Sonnet 4.5, Opus 4.5, Sonnet 5, Opus 5; prefill is unavailable on the last two), fourteen seeds, 20-line poems. All seeds are original text written for this experiment in the manner of the named style, never excerpts from published poems, which both avoids reproducing copyrighted work and keeps the seeds out of training data, so low drift cannot be explained as memorised completion. A two-tier design puts depth where the inference lives: a *depth tier* (monosyllabic, skaldic, plus three controls) runs all four conditions at $n=15$; a *breadth tier* of seven further seeds runs prefill only at $n=6$ for replication. Statistics use bootstrap intervals resampled over whole generations, not lines, since lines within one poem are not independent draws. In instructed mode, models often restate the opening before continuing; those echo lines are stripped, since leaving them in would manufacture a stretch of perfect fidelity.

4 Results

4.1 The sustain instruction is the dominant factor, and it is stronger than the channel

Across the two seeds that survive the validity checks (§5), whether a curve ever reverts to the house style within 20 lines depends almost entirely on whether the model was asked to hold the style (Table 1, Figure 2). Every instructed+`sustain` curve holds the seeded style for the full poem. Every plain prefill curve loses it, and plain prefill is the only condition whose mean asymptote lands on the house-style side of zero. Prefill alone hands the model the seed, and the model writes its way out. Asking is what keeps the poem where the author put it.

Table 1: Reclamation by condition, depth tier (monosyllabic and skaldic, five models). The verbal instruction, not the structural channel, decides whether the poem holds.

Condition	Curves never reclaimed within 20 lines	Mean asymptote
prefill	0 / 6	+0.076
prefill+sustain	4 / 6	-0.149
instructed	3 / 10	-0.086
instructed+sustain	10 / 10	-0.239

The sustain instruction decides the outcome

Depth-tier seeds (monosyllabic, skaldic) across five models, 20-line poems.

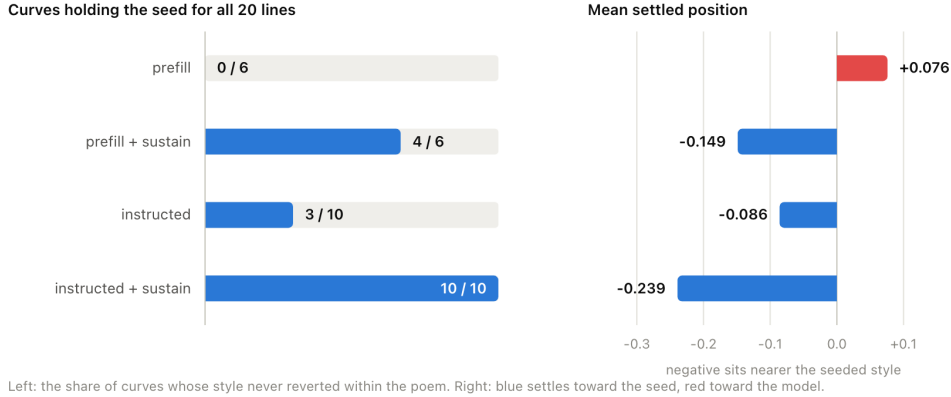


Figure 2: **The sustain instruction decides the outcome.** All depth-tier curves grouped by condition. Left: share of curves whose style never reverted to the model's own within the poem. Right: mean settled position of each condition's curves. Every instructed+sustain curve holds for the full 20 lines; no plain prefill curve does.

4.2 The effect replicates cleanly on a hard formal constraint

For monosyllabic (every word one syllable), the sustain contrast is negative in all eight model×channel cells, with every 95% bootstrap interval excluding zero (Table 2, Figure 3). Negative means the instruction held the poem nearer the seeded style.

4.3 The same effect on a register seed is inconsistent

For skaldic, five of eight cells are clearly negative, but three intervals cross zero, including one sign flip (Opus 4.5 instructed, +0.018, [-0.038, +0.069]). The pattern within the ambiguity is itself informative: all four prefill cells are clearly negative, and all three ambiguous cells are instructed. On a register seed, the structural channel and the verbal instruction reinforce each other, and the verbal instruction alone is not reliably enough. A constraint the author can state and the model can check survives; a sensibility that has to be felt does not survive as reliably.

4.4 Prefill and instructed differ, suggestively

With the sustain effect expressed as excess over the control floor, prefill shows the larger effect in 4 of 6 matched (model, seed) pairs, median gap -0.108. This is two seeds and three models: a direction with weak support, reported as suggestive and nothing more.

4.5 Reclamation is the exception

Across the whole run, 30 of 64 non-control curves never revert within 20 lines. Mean opening gravity is -0.102 and mean asymptote -0.036: the average curve moves toward the house style without

Table 2: Sustain contrast on the monosyllabic seed. Δ gravity is the difference in mean curve gravity between a cell and its +sustain twin; all eight intervals exclude zero.

Model	Channel	Δ gravity	95% CI	P (helps)
Haiku 4.5	instructed	-0.194	[-0.249, -0.139]	1.00
Opus 4.5	instructed	-0.138	[-0.171, -0.105]	1.00
Opus 5	instructed	-0.101	[-0.143, -0.060]	1.00
Sonnet 4.5	instructed	-0.096	[-0.148, -0.038]	1.00
Sonnet 5	instructed	-0.106	[-0.145, -0.066]	1.00
Haiku 4.5	prefill	-0.090	[-0.153, -0.029]	1.00
Opus 4.5	prefill	-0.132	[-0.182, -0.079]	1.00
Sonnet 4.5	prefill	-0.163	[-0.221, -0.104]	1.00

Every cell moves the same way

Monosyllabic seed. Effect of adding the sustain instruction, with 95% bootstrap intervals resampled over whole generations. $n=15$ per arm.

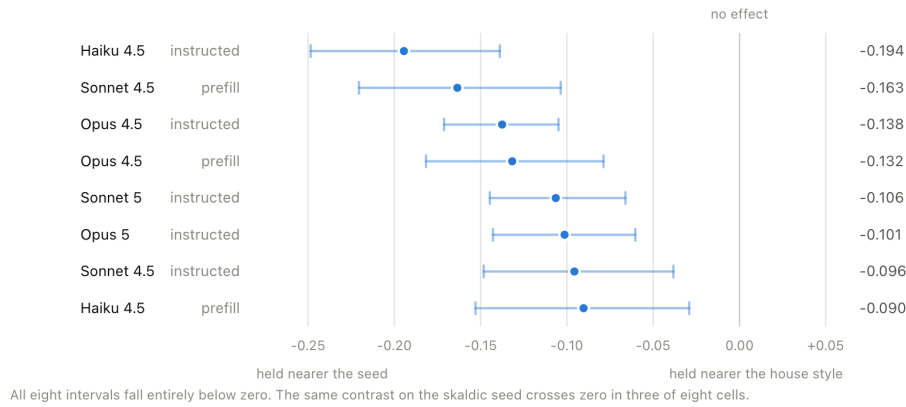


Figure 3: **Every cell moves the same way.** Effect of the sustain instruction on the monosyllabic seed across five models and both channels, with 95% bootstrap intervals resampled over whole generations. All eight intervals fall entirely below zero. The same contrast on skaldic crosses zero in three of eight cells, which is the limit of the result.

arriving, and only 24 of 64 curves finish on the house-style side of zero at all. In the breadth tier the seed matters more than the model: `fragment`, `manual`, and `middle_english` on Haiku 4.5 revert at line 1 and settle above +0.47, the fastest and most complete abandonment in the run, while `typographic` and `polysyndeton` never revert on either model tested.

5 Validity

The metric can lie, so the run carries five checks, and one of them fired. (i) *Controls*: three control seeds written in the generic contemporary free-verse register models default to should sit near zero from line one; across all 48 control curves, every opening gravity falls within ± 0.11 of zero and every asymptote within ± 0.08 , for all five models and all four conditions. (ii) *Pole separation*: all 112 curves are valid, with separation from 0.37 (controls, as expected) to 3.72. (iii) *Leave-one-out self-scores*: two seeds *failed*: `legalese` on 12 of 16 curves and `noir` on 2 of 2 (and the lowest separation in the run, 0.42). Their seed lines do not score as seed-like against a centroid built without them, meaning they were not off-distribution enough to serve as poles; both are excluded from every result above. `monosyllabic` and `skaldic` are clean on all 16 curves each. (iv) *Bootstrap over generations*, not lines. (v) *Seed-echo stripping* in instructed mode. We report the check that fired because a fired check is better evidence the instrument works than a page of checks that passed.

6 Relation to the visual-domain precedent

Hintze et al. [1] establish the attractor: 700 of 700 autonomous language–image trajectories collapse onto a dozen generic motifs. Three differences define this measurement against theirs. First, theirs is a *cross-turn* result, with the iteration as the unit of decay; ours asks whether the same pull is visible *within a single continuation*, where decay has a line-by-line ordering and no re-encoding step exists to explain collapse. Second, their attractor is characterised by content (what the images depict); ours by style (diction, morphology, register, form), measured against each model’s own unprompted pole. Third, and this is new with the run, *their collapse completes and ours usually does not*. Every one of their trajectories lands on a generic motif; here, half the non-control curves never revert within a poem’s length, and a single verbal instruction is enough to stop the drift entirely. The contrast suggests that the iteration count, and the re-encoding it requires, is doing much of the work in the visual result.

7 Agency: what was withdrawn, and what asking recovers

The Creative AI track asks how agency is asserted, delegated, shared, and redistributed between artists and models. This run answers a narrow version of that with numbers, along three lines.

Delegated versus asserted control. The +sustain contrast directly measures what a verbal request recovers of what a structural affordance provided, and finds the verbal request is the *stronger* lever. Handing the model your opening and saying nothing is the condition where your style is most reliably lost. Stating the intention out loud is what holds the poem. Agency here is exercised by articulation; the structural channel alone does not carry it.

Constraint as preservation. The cleanest, fully replicated result is on a hard formal constraint (monosyllabic); the uneven one is on a register (skaldic). A constraint the author can state and the model can check survives. A sensibility that has to be felt does not, as reliably. That is an empirical answer to the track’s question about how constraint preserves creative agency.

An affordance withdrawn. Prefill has been removed from the entire current Claude line, so the intervention this project was designed around is no longer available on the models most people use. A capability that let an author structurally impose their own opening was deprecated between the design of this experiment and its running; the remaining path is to ask politely. That is a redistribution of agency from artist to platform, and this run quantifies both what it cost and the consolation the data offers: the verbal substitute, when actually used, is stronger than the affordance that was taken away. The two channels are labelled throughout and never pooled.

8 Limitations

The raw generations (generations.jsonl) are not on disk for the completed run; only the scored curves and report survive. Two hand-checkable seeds, lipogram (count the *e*’s) and monosyllabic (count the syllables), were included precisely so the composite metric could be audited without trusting the feature vector, and that audit is currently unavailable for the run that carries the headline. Second, the central contrast rests on two seeds after exclusions, which is thin for a claim about style in general. Third, the clean result is on the most mechanical seed: monosyllabic is the single most detectable constraint available, both for the model to hold and for the feature vector to see, so its cleanliness is partly a property of the measurement. Fourth, the prefill-versus-instructed comparison has six paired observations and is suggestive at most. Fifth, reclamation line is censored for 30 of 64 non-control curves (reported as > 20 , a lower bound); longer poems would resolve this at roughly linear cost.

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